Bias Assessment Report

Paper: Multi-Species Synbiotic Supplementation After Antibiotics Promotes Recovery of Microbial Diversity and Function, and Increases Gut Barrier Integrity: A Randomized, Placebo-Controlled Trial
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Verification: Enabled

Overall Assessment

Metric	Value
Severity	**HIGH**
Bias Probability	82%
Confidence	HIGH

Reasoning: Step-by-step reasoning across all five domains:

1. STATISTICAL REPORTING (HIGH): Effect sizes are reported almost exclusively as fold-changes and percentage differences without any absolute values in either arm. The authors themselves acknowledge that extreme fold-changes (e.g., 13,008% for UroA, 2587-fold for strain abundance) are mathematical artifacts of near-zero denominators in the placebo arm, yet continue to use these as primary effect descriptors. No ARR, NNT, or raw event rates are provided. A post-hoc subgroup (low-baseline butyrate) is used to generate the butyrate finding without pre-specification or multiplicity correction. No correction for multiple comparisons is applied across numerous endpoints and timepoints. This pattern — relative-only reporting combined with acknowledged artifact inflation and subgroup emphasis — constitutes HIGH severity.

2. SPIN (HIGH): The conclusions assert definitive clinical benefit ('promotes recovery,' 'accelerated restoration,' 'compelling evidence,' 'distinct advantages') from exclusively surrogate/mechanistic endpoints in a small healthy-volunteer trial. No patient-centred outcomes were collected, yet the conclusions link findings to antibiotic-associated GI symptoms, IBS, cardiovascular and renal disease. The absence of patient-reported outcomes is acknowledged only as a minor limitation rather than a fundamental constraint. The title itself claims clinical benefits ('promotes recovery,' 'increases gut barrier integrity') that surrogates cannot confirm. The conclusions contain no uncertainty

language, no recommendation for further trials, and no acknowledgment of the mechanistic-only nature of the evidence. This meets HIGH spin criteria.

3. OUTCOME REPORTING (MODERATE): All primary endpoints are surrogate biomarkers (microbiome diversity indices, SCFAs, UroA, pCS, TMAO, lactulose permeability ratio) without established validation as predictors of patient-centred outcomes. The study is explicitly framed as 'proof-of-mechanism' but conclusions extend to clinical relevance. The post-hoc butyrate subgroup analysis is presented without pre-specification. While surrogate endpoints are appropriate for mechanistic studies, the clinical extrapolation without validation linkage elevates this to MODERATE rather than LOW.

4. CONFLICT OF INTEREST (HIGH): The study is entirely industry-funded by Seed Health, Inc. (manufacturer of DS-01). Two authors are employees and shareholders of the sponsor; two additional authors are paid advisors. Critically, the sponsor employee (B.A.N.) performed all data analysis and led manuscript drafting, despite the claim that clinical operations were independent. COI is disclosed at the author level (mitigating from CRITICAL), but the combination of full industry funding, multiple author-level financial conflicts, and sponsor-employee control of data analysis and manuscript preparation represents a HIGH severity concern. The abstract does not disclose funding or COI.

5. METHODOLOGY (HIGH): The most serious concern is per-protocol-only analysis with 34.4% overall attrition and differential dropout (25% placebo vs. 43.75% treatment arm) from a small trial (n=32 randomized, n=21 analyzed). No ITT or mITT analysis is reported. Differential dropout in the treatment arm could introduce selection bias favoring completers who tolerated or responded to the intervention. The sample size of 21 completers is very small for the number of complex endpoints tested. No multiple comparisons correction is applied. The healthy-volunteer antibiotic challenge model limits generalizability to actual patients. The dual-threshold colonization detection approach with post-hoc selection of the lower-threshold method raises analytical flexibility concerns. The combination of per-protocol-only analysis with high differential attrition and no multiplicity correction constitutes HIGH severity.

Overall: Four of five domains reach HIGH severity, with the fifth (outcome reporting) at MODERATE. The pattern across domains — industry employees controlling data analysis and writing, relative-only reporting of artifact-inflated fold-changes, per-protocol analysis with differential attrition, surrogate-only endpoints with clinical extrapolation, and spin in conclusions — is consistent with systematic bias favoring the sponsor's commercial product. Overall severity is HIGH with bias probability 0.82.

Statistical Reporting

Severity: HIGH

Flag	Value
Relative Only	Yes
Absolute Reported	No
Nnt Reported	No
Baseline Risk Reported	No
Selective P Values	No
Subgroup Emphasis	Yes

Evidence:

> fecal butyrate (119%, $p < 0.05$), fecal acetate (62%, $p < 0.01$), and UroA (13,008%, $p < 0.05$), whereas detrimental metabolite pCS decreased (68%, $p < 0.05$) compared to placebo

> gut barrier integrity rapidly (Day 7; 305%, $p < 0.05$) and over the long-term (Day 91; 161%, $p < 0.05$)

> Day 7, 2587-fold-change, $p < 0.0001$... Day 14, 118-fold-change, $p < 0.001$... Day 49, 268-fold-change, $p < 0.0001$... Day 91, 435-fold-change, $p < 0.0001$

> Notably, these high fold-changes reflect the low level of these strains in the placebo arm and the high precision of the detection method

> Significance was defined a priori at $p < 0.05$ [with no mention of multiple comparisons correction across numerous endpoints and timepoints]

> Thus, we evaluated the recovery of butyrate production after antibiotics in a population with low-baseline fecal butyrate [post-hoc subgroup]

Spin

Severity: HIGH

Flag	Value
Spin Level	high
Conclusion Matches Results	No
Causal Language From Observational	No
Focus On Secondary When Primary Ns	No
Inappropriate Extrapolation	Yes
Title Spin	Yes

Evidence:

> During and after antibiotics, this multi-species synbiotic promotes recovery of gut microbiome diversity and native beneficial microbes, microbiome metabolite recovery, and gut barrier function, all of which underpin antibiotic-associated gastrointestinal symptoms.

> the data from this trial provide compelling evidence that a 24-strain synbiotic supports the preservation and recovery of microbiome community structure

> In contrast to previous studies that question the utility and raised concerns about delays in microbiome recovery due to probiotic use, this trial suggests that this synbiotic may offer distinct advantages.

> administration of a multi-species synbiotic during and following broad-spectrum antibiotic exposure accelerated restoration of microbiome diversity, promoted recovery of key microbial metabolic outputs, and improved indices of gut barrier function

> This trial focused on mechanistic endpoints and did not include patient-reported GI symptom outcomes. While the observed improvements in microbiome composition, metabolite recovery, and gut barrier integrity provide strong biological plausibility for clinical benefit...

> Recently, UMGS1312 sp900550625 was reported to be associated with protection against irritable bowel syndrome (IBS), sarcopenia, stress, and metabolic conditions... Importantly, the multi-species synbiotic significantly increased the recovery of the novel beneficial native species UMGS1312 sp900550625 after antibiotics compared to placebo.

Outcome Reporting

Severity: MODERATE

Flag	Value
Primary Outcome Type	surrogate
Surrogate Without Validation	Yes
Composite Not Disaggregated	No

Evidence:

> Endpoints included fecal microbiome composition, fecal acetate and butyrate levels, urinary Urolithin A (UroA), serum p-cresol sulfate (pCS), gut barrier integrity, and safety.

> The study endpoints included microbiome compositional dynamics assessed by whole-genome shotgun sequencing, fecal SCFAs, urinary UroA production, serum pCS and TMAO, gut barrier integrity assessed by a lactulose-based test, and safety parameters.

> accelerated restoration of microbiome diversity, promoted recovery of key microbial metabolic outputs, and improved indices of gut barrier function, outcomes mechanistically linked to post-antibiotic recovery

> This trial focused on mechanistic endpoints and did not include patient-reported GI symptom outcomes.

> Thus, we evaluated the recovery of butyrate production after antibiotics in a population with low-baseline fecal butyrate [post-hoc subgroup without pre-specification disclosure]

Conflict of Interest

Severity: HIGH

Flag	Value
Funding Type	industry
Funding Disclosed In Abstract	No
Industry Author Affiliations	Yes
Coi Disclosed	Yes

Evidence:

> B.A.N. and Z.K. are employees and shareholders of Seed Health, Inc. G.R. and R.J. serve as advisors to Seed Health, Inc.

> Conceptualization, B.A.N., G.R. and Z.K.; data analysis B.A.N.; writing—preparation of the original draft, B.A.N., Z.K. and R.J.

> The sponsor (Seed Health, Inc.) had no role in clinical trial operations, which were performed independently by KGK Science, Inc... data analysis B.A.N.

> DS-01, Seed Health, Inc., Venice, CA, USA... the consortium is engineered to drive ecological recovery via network interactions.

Methodology

Severity: HIGH

Flag	Value
Inappropriate Comparator	No
Enrichment Design	No
Per Protocol Only	Yes
Premature Stopping	No
Short Follow Up	No

Evidence:

> Eleven participants were lost to follow-up (Placebo: n = 4; DS-01: n = 7), leaving 21 participants who completed the trial and were included in the statistical analysis.

> Of the 86 individuals who responded to recruitment and underwent eligibility screening, 32 met the inclusion criteria... leaving 21 participants who completed the trial and were included in the statistical analysis.

> healthy adult participants received a daily synbiotic... All participants also received ciprofloxacin (500 mg orally twice daily) and metronidazole (500 mg orally three times daily) for the first 7 days.

> participants in this study were healthy adults without baseline GI or underlying clinical conditions, so the findings may not be directly generalizable to populations at the highest risk for antibiotic-associated complications

> L. rhamnosus was significantly increased at Day 7 (36-fold-change, $p < 0.05$), Day 49 (74-fold-change, $p < 0.01$), and Day 91 (75-fold-change, $p < 0.05$)... L. plantarum was significantly increased at Day 7 (1337-fold-change, $p < 0.001$), Day 14 (100-fold-change, $p < 0.01$), Day 49 (442-fold-change, $p < 0.001$), and Day 91 (298-fold-change, $p < 0.01$) [no multiple comparisons correction]

> Synbiotic strain abundance was observed two ways... For approach (2), reported in the results, synbiotic strain abundance was assessed alongside >600 additional genomes... A synbiotic strain genome was considered present if ≥15% unique and ≥25% total k-mers were detected

Recommended Verification Steps

- Check ClinicalTrials.gov registration (search by DS-01, Seed Health, ciprofloxacin synbiotic) to verify pre-specified primary outcomes, sample size justification, and whether the butyrate subgroup analysis was pre-registered.
- Request raw data or absolute values in both arms for all primary endpoints to enable independent assessment of clinical significance beyond relative measures.
- Verify whether an ITT or mITT analysis was conducted and why it was not reported as the primary analysis given 34.4% attrition with differential dropout.
- Assess whether multiple comparisons correction was pre-specified in the statistical analysis plan and why it was not applied given the large number of simultaneous hypothesis tests.
- Examine the financial relationships of all four conflicted authors (B.A.N., Z.K., G.R., R.J.) with Seed Health, Inc. including equity holdings, consulting fees, and advisory compensation.
- Evaluate the ethical approval documentation for the healthy-volunteer antibiotic challenge design to confirm IRB review of the risk-benefit ratio of administering broad-spectrum antibiotics to healthy participants.
- Assess the pre-registration status of the dual-threshold colonization detection approach (approach 1 vs. approach 2) to determine whether the lower-threshold method was selected post-hoc.
- Conduct a systematic review or meta-analysis of all Seed Health-funded synbiotic trials to assess publication bias and consistency of effect sizes across independent replications.
- Verify whether the referenced prior validation studies (references 32 and 33) were also funded by Seed Health, Inc. to assess the independence of the supporting evidence base.

Verification Results

clinicaltrials_gov [Pass]

Trial NCT04171466 — Sponsor: Seed Health (INDUSTRY)

Trial: Metagenomic and Metabolomic Reconstitution of Gut Microbiota After Broad Spectrum Antibiotic Therapy
NCT ID: NCT04171466
Status: COMPLETED
Phase: NA
Sponsor: Seed Health (INDUSTRY)
Funding source: industry
Registered primary outcomes: Change in microbiota composition at 3 months as assessed by whole genome shotgun sequencing.
Registered secondary outcomes: Difference in serum LPS-binding protein (LBP) at Day

7.; Difference in the Intestinal Permeability Assessment (IPA) at Day 7 as measured by Lactulose/mannitol testing.; Metabolomic profile of stool samples.; Number of participants with improved Antibiotic-Associated Gastrointestinal Function

No outcome switching detected.

open_payments [Pass]

No payment records found in CMS Open Payments.

Searched for: Brooke A. Napier, Jessica R. Allegretti, Paul Feuerstadt. Note: CMS Open Payments covers US physicians only.

orcid [Pass]

Found 3 affiliations

- **Brooke A. Napier**: Seed Health (None, -)
- **Paul Feuerstadt**: Gastroenterology Center of Connecticut (Attending Gastroenterologist, 2011-)
- **Paul Feuerstadt**: Danbury Hospital (Attending Physician, 2010-2011)

effect_size_audit [Pass]

Pattern: no_effect_size (score: 0.00)

Reporting pattern: no_effect_size
Reporting bias score: 0.00

europmc [Fail]

No PMID provided for Europe PMC lookup.

retraction_watch [Pass]

No retraction notices found for DOI 10.3390/antibiotics15020138.

Crossref: no retraction or correction notices.
No retraction, withdrawal, or expression of concern detected.

Refined Assessment (Post-Verification)

Metric	Initial	Refined
Severity	HIGH	**HIGH**
Bias Probability	82%	**85%**

Refined Reasoning: Verification results from ClinicalTrials.gov (NCT04171466) confirm and in one important respect worsen the initial assessment:

1. STATISTICAL REPORTING (HIGH, unchanged): Verification confirms relative-only reporting throughout. The effect size audit tool returned 'no_effect_size' with a bias score of 0.00, which reflects the tool's inability to extract absolute values — consistent with the finding that none are reported. The acknowledged artifact inflation of fold-changes (near-zero denominators) and absence of multiple comparisons correction remain unaddressed. Severity maintained at HIGH.

2. SPIN (HIGH, unchanged): The ClinicalTrials.gov registration confirms the trial was registered with a patient-reported GI symptom outcome ('Number of participants with improved Antibiotic-Associated Gastrointestinal Function') as a secondary endpoint. The abstract's conclusions extrapolate to clinical GI symptom benefit while omitting results for this registered patient-reported outcome — a pattern consistent with selective emphasis on favorable surrogate findings. This reinforces rather than mitigates the spin assessment. Severity maintained at HIGH.

3. OUTCOME REPORTING (MODERATE, confidence increased): The ClinicalTrials.gov registration reveals a critical new finding: a registered secondary outcome ('Number of participants with improved Antibiotic-Associated Gastrointestinal Function') is entirely absent from the abstract. This is a patient-reported clinical outcome that was pre-specified but not reported, while exclusively surrogate mechanistic endpoints are presented. This selective non-reporting of a pre-registered patient-centred outcome is a significant additional concern. No outcome switching was detected (the registered primary outcome — microbiota composition by WGS — is reported), but the omission of the GI symptom secondary outcome elevates concern. Severity maintained at MODERATE but confidence in this rating increases substantially; the finding approaches HIGH given the selective omission of the only patient-reported endpoint.

4. CONFLICT OF INTEREST (HIGH, unchanged): ClinicalTrials.gov confirms Seed Health as the sole sponsor with no independent co-sponsor, corroborating full industry funding. ORCID confirms B.A.N.'s affiliation with Seed Health. CMS Open Payments returned no data for the authors searched, but this tool covers US physicians only and the primary conflicted authors (B.A.N., Z.K.) are industry employees rather than physicians, so the null result is uninformative rather than exculpatory. No retraction or correction notices were found. Severity maintained at HIGH.

5. METHODOLOGY (HIGH, confidence increased): The ClinicalTrials.gov registration confirms the pre-specified primary outcome was microbiota composition at 3 months — consistent with what is reported. However, the registration of 'Number of participants with improved Antibiotic-Associated Gastrointestinal Function' as a secondary outcome, combined with its complete absence from the abstract, raises a selective reporting concern beyond the per-protocol-only analysis and differential attrition already identified. The 34.4% overall attrition with differential dropout (25% placebo vs. 43.75% treatment) and absence of ITT analysis remain unaddressed by verification. Severity maintained at HIGH.

Overall: Verification confirms all five initial domain assessments and adds a materially important new finding — a pre-registered patient-reported GI symptom outcome is absent from the abstract while the conclusions claim clinical relevance for GI symptoms. This selective non-reporting of the only patient-centred endpoint, combined with the confirmed full industry sponsorship, sponsor-employee data analysis, per-protocol-only analysis with differential attrition, relative-only reporting of artifact-inflated fold-changes, and high spin in conclusions, warrants a slight upward revision of overall bias probability from 0.82 to 0.85. Overall severity remains HIGH.

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